

Task 12: Session 12 (AIML) - Assignment Questions

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Domain: AIML

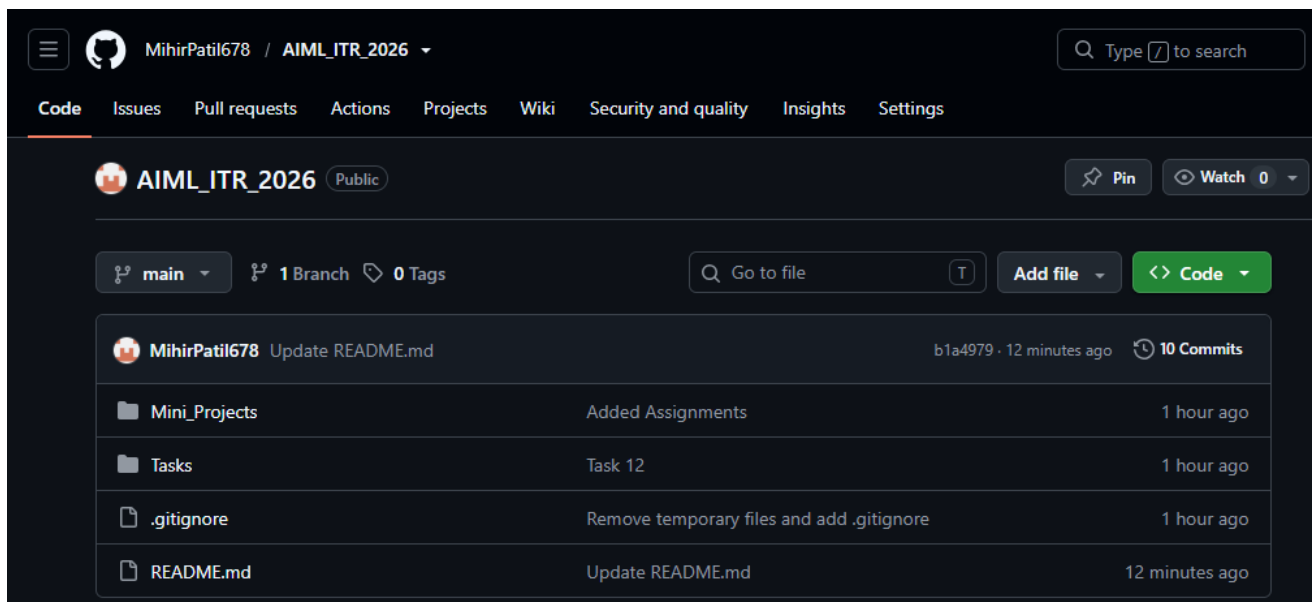
College & Branch Name: Zeal Polytechnic, Diploma In AIML

GitHub: https://github.com/MihirPatil678/AIML_ITR_2026/tree/main/Tasks/Task_12

1. GitHub Account Setup

- Create a GitHub account (if you don't have one).
- Create a new repository named Python-Basic-Projects.
- After implementing each feature in each question's solution, commit your code with a proper commit message (e.g., "Implement Loop for Counting Vowels" or "Create analyze_string function").
- After completing the full project, push everything to GitHub.

Output:



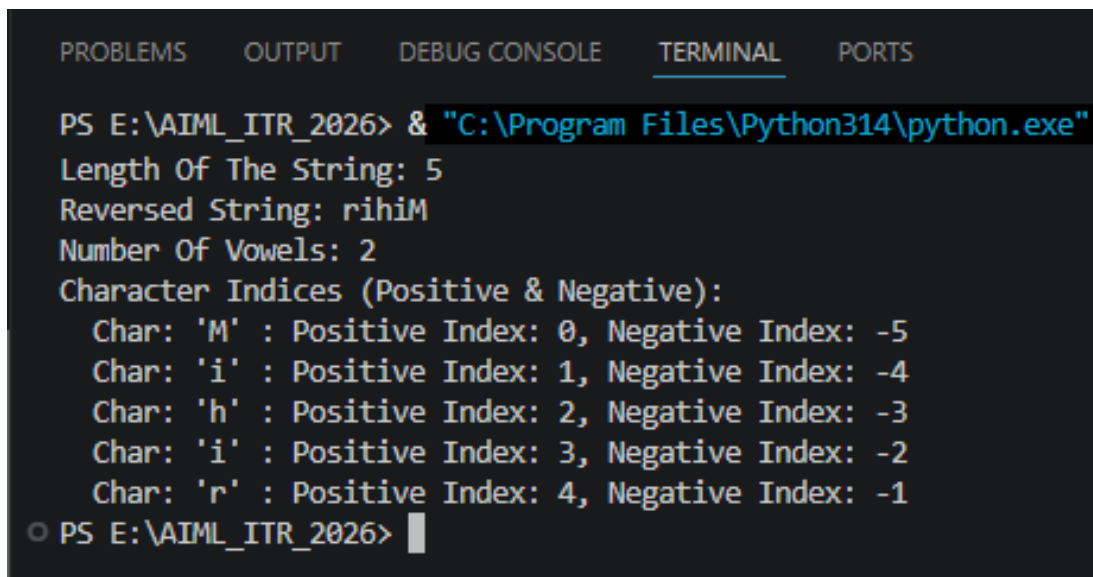
2. Strings + Loops + Functions

Write a function `analyze_string(s)` that takes a string as input and:

- Prints its length using `len()`.
- Prints the string in reverse using slicing.
- Counts and prints how many vowels (a,e,i,o,u) are in the string (case insensitive).
- Uses a for loop with `range()` to print each character with its positive and negative index.

Call this function with user input and handle empty string case.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe"
Length Of The String: 5
Reversed String: rihiM
Number Of Vowels: 2
Character Indices (Positive & Negative):
  Char: 'M' : Positive Index: 0, Negative Index: -5
  Char: 'i' : Positive Index: 1, Negative Index: -4
  Char: 'h' : Positive Index: 2, Negative Index: -3
  Char: 'i' : Positive Index: 3, Negative Index: -2
  Char: 'r' : Positive Index: 4, Negative Index: -1
PS E:\AIML_ITR_2026> |
```

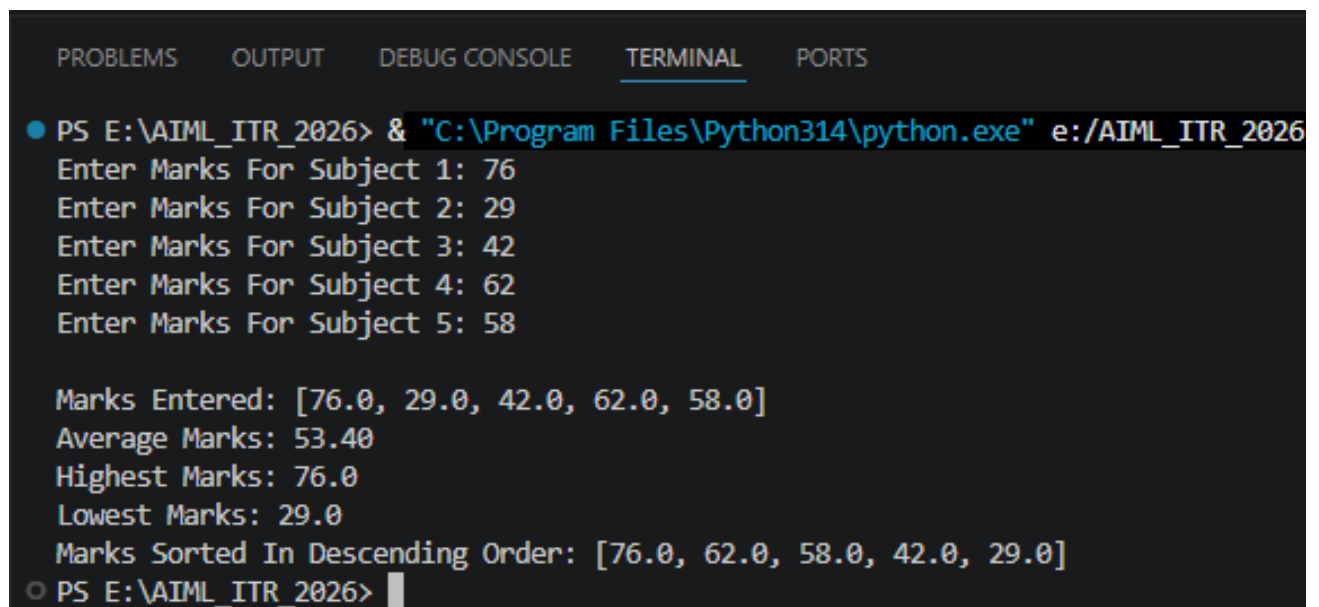
3. Lists + Functions + Exception Handling

Create a function `manage_marks()` that:

- Takes 5 subject marks as input from the user.
- Stores them in a list.
- Handles `ValueError` if non-numeric input is given.
- Calculates and prints the average, highest, and lowest marks.
- Sorts the list in descending order and prints it.

Use meaningful comments.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe" e:/AIML_ITR_2026
Enter Marks For Subject 1: 76
Enter Marks For Subject 2: 29
Enter Marks For Subject 3: 42
Enter Marks For Subject 4: 62
Enter Marks For Subject 5: 58

Marks Entered: [76.0, 29.0, 42.0, 62.0, 58.0]
Average Marks: 53.40
Highest Marks: 76.0
Lowest Marks: 29.0
Marks Sorted In Descending Order: [76.0, 62.0, 58.0, 42.0, 29.0]
○ PS E:\AIML_ITR_2026> 
```

4. OOP + Lists + Exception Handling

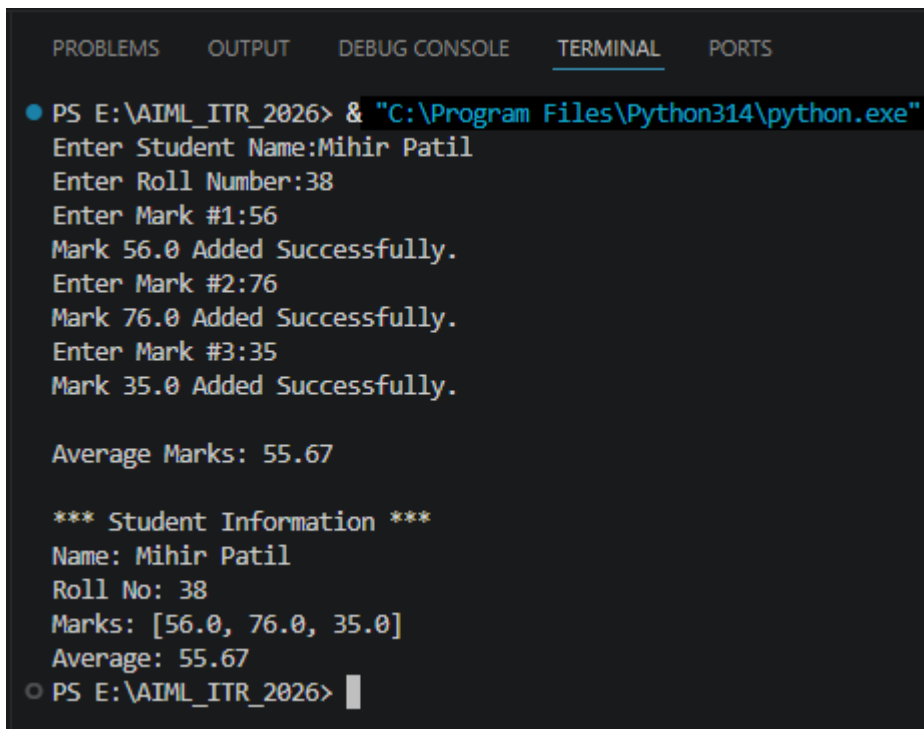
Create a class Student with attributes: name, roll_no, and marks_list (a list of marks).

Include methods:

- `__init__` to initialize the student.
- `add_mark(mark)` to add a mark to the list (handle invalid marks).
- `get_average()` to return the average.
- `display_info()` to show all details.

Create one student object, add marks using user input, and demonstrate all methods with exception handling.

Output:



```
PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter Student Name:Mihir Patil
Enter Roll Number:38
Enter Mark #1:56
Mark 56.0 Added Successfully.
Enter Mark #2:76
Mark 76.0 Added Successfully.
Enter Mark #3:35
Mark 35.0 Added Successfully.

Average Marks: 55.67

*** Student Information ***
Name: Mihir Patil
Roll No: 38
Marks: [56.0, 76.0, 35.0]
Average: 55.67
PS E:\AIML_ITR_2026> 
```

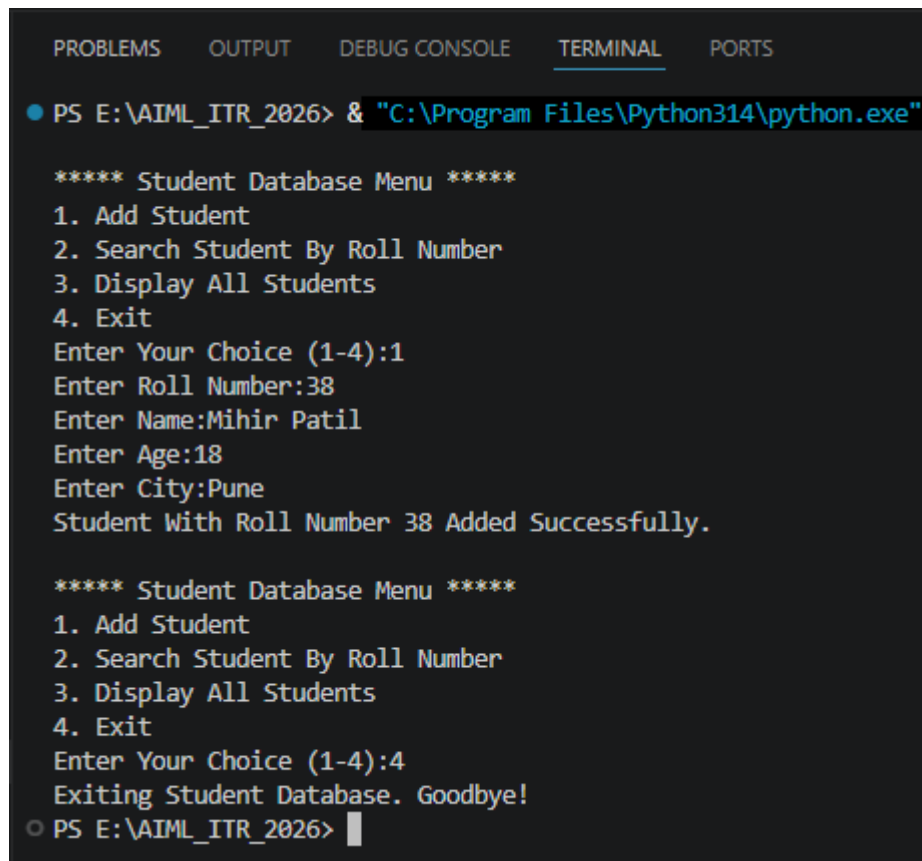
5. Dictionaries + Functions + Control Structures

Write a function `student_database()` that uses a dictionary (roll number as key) to store student records.

- Provide a menu using while loop:
- Add student (name, age, city)
- Search student by roll number
- Display all students
- Exit

Use `get()`, `update()`, and proper exception handling for invalid inputs.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe"
**** Student Database Menu ****
1. Add Student
2. Search Student By Roll Number
3. Display All Students
4. Exit
Enter Your Choice (1-4):1
Enter Roll Number:38
Enter Name:Mihir Patil
Enter Age:18
Enter City:Pune
Student With Roll Number 38 Added Successfully.

**** Student Database Menu ****
1. Add Student
2. Search Student By Roll Number
3. Display All Students
4. Exit
Enter Your Choice (1-4):4
Exiting Student Database. Goodbye!
○ PS E:\AIML_ITR_2026> |
```

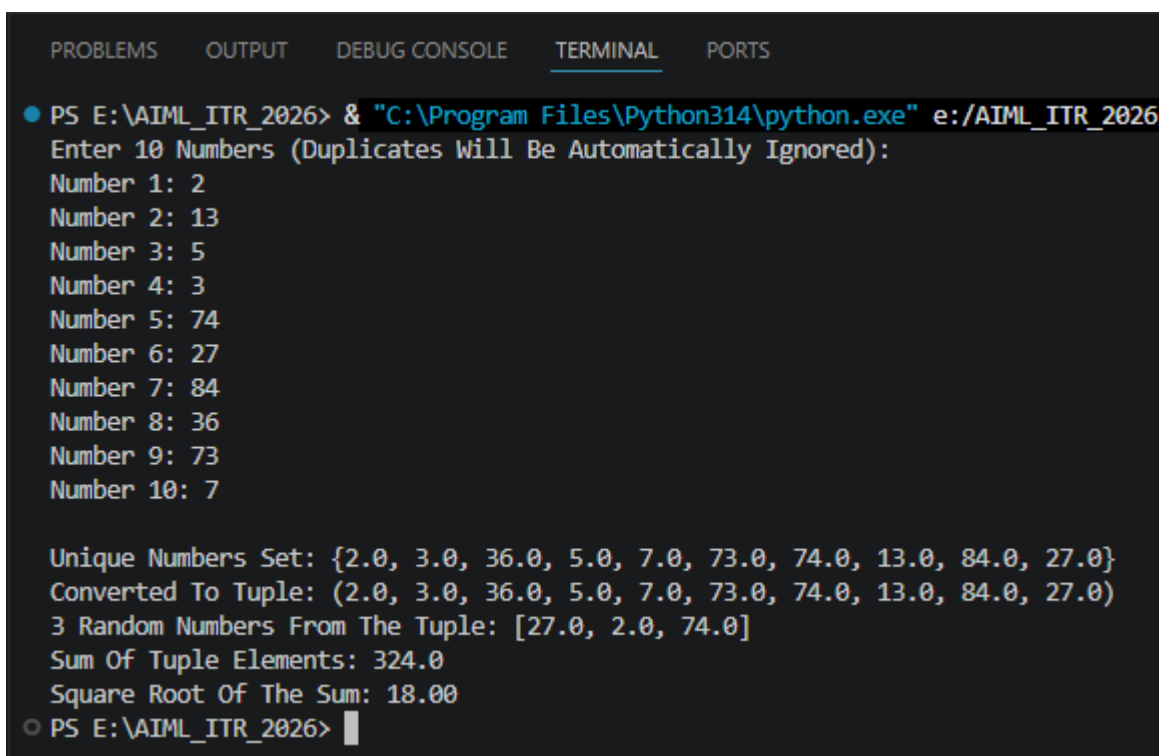
6. Sets + Tuples + Modules

Write a program that:

- Takes 10 numbers as input and stores unique numbers in a set.
- Converts that set into a tuple.
- Uses random module to pick and print 3 random numbers from the tuple.
- Uses math module to print the square root of the sum of the tuple elements.

Handle possible exceptions.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe" e:/AIML_ITR_2026
Enter 10 Numbers (Duplicates Will Be Automatically Ignored):
Number 1: 2
Number 2: 13
Number 3: 5
Number 4: 3
Number 5: 74
Number 6: 27
Number 7: 84
Number 8: 36
Number 9: 73
Number 10: 7

Unique Numbers Set: {2.0, 3.0, 36.0, 5.0, 7.0, 73.0, 74.0, 13.0, 84.0, 27.0}
Converted To Tuple: (2.0, 3.0, 36.0, 5.0, 7.0, 73.0, 74.0, 13.0, 84.0, 27.0)
3 Random Numbers From The Tuple: [27.0, 2.0, 74.0]
Sum Of Tuple Elements: 324.0
Square Root Of The Sum: 18.00
○ PS E:\AIML_ITR_2026> |
```

7. Lambda + range() + Lists + Exception Handling

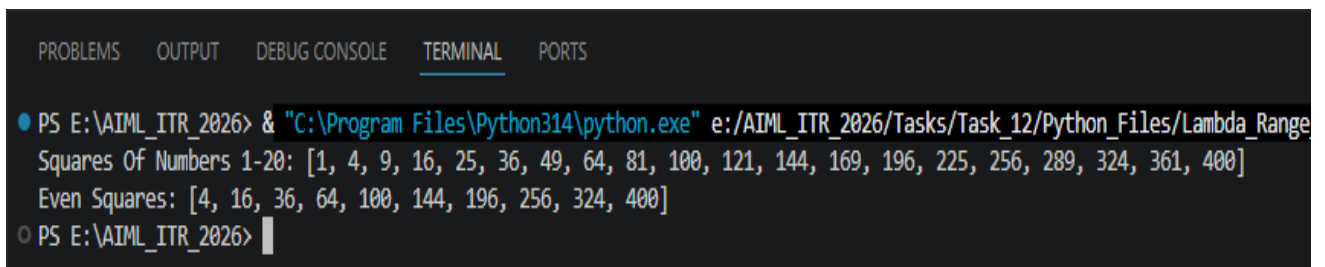
Create a lambda function to calculate the square of a number.

Write a program that:

- Uses range(1, 21) to generate numbers.
- Stores the squares of these numbers in a list using the lambda function.
- Prints only the even squares from the list.

Add exception handling if needed.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe" e:/AIML_ITR_2026/Tasks/Task_12/Python_Files/Lambda_Range
Squares Of Numbers 1-20: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400]
Even Squares: [4, 16, 36, 64, 100, 144, 196, 256, 324, 400]
○ PS E:\AIML_ITR_2026> |
```

8. Tuples + Dictionaries + OOP

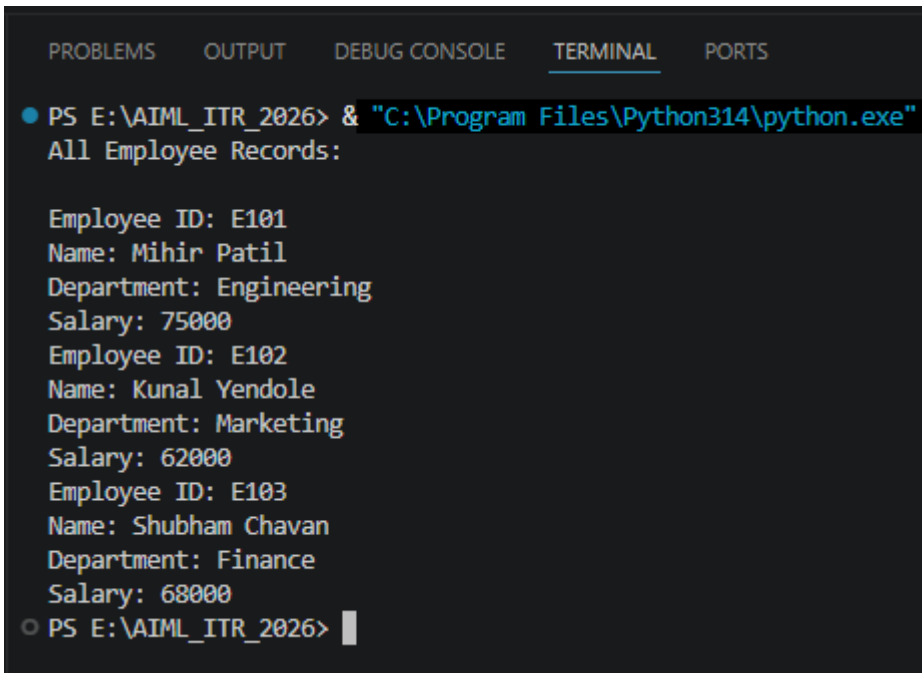
Create a class Employee with:

- Attributes: emp_id, name, details (a tuple containing department and salary).
- Method show_details() to print all information.

Create a dictionary where employee ID is the key and Employee object is the value.

Add 3 employees and display all using a loop.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe"
All Employee Records:

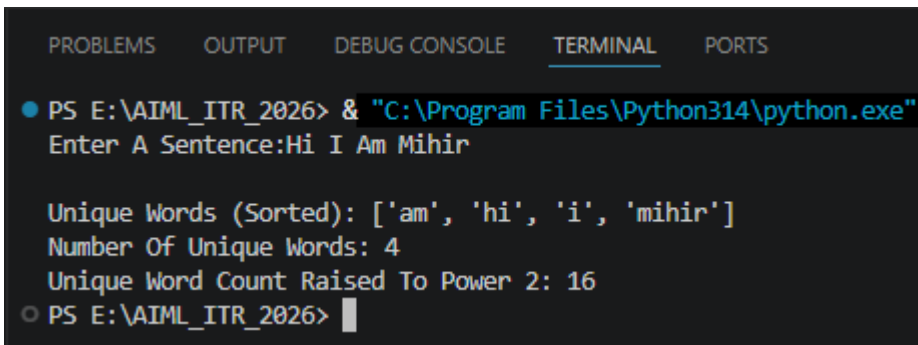
Employee ID: E101
Name: Mihir Patil
Department: Engineering
Salary: 75000
Employee ID: E102
Name: Kunal Yendole
Department: Marketing
Salary: 62000
Employee ID: E103
Name: Shubham Chavan
Department: Finance
Salary: 68000
○ PS E:\AIML_ITR_2026> |
```


9. Strings + Sets + Exception Handling + Modules

Write a program that:

- Takes a sentence as input.
- Extracts all unique words (using set) from the sentence.
- Prints the unique words in sorted order.
- Uses math module to print the total number of unique words raised to power 2.
- Handle any possible exceptions gracefully.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A Sentence:Hi I Am Mihir

Unique Words (Sorted): ['am', 'hi', 'i', 'mihir']
Number Of Unique Words: 4
Unique Word Count Raised To Power 2: 16
○ PS E:\AIML_ITR_2026> 
```

10. Mini Project - Comprehensive

Create a Smart Calculator & Data Manager program that combines multiple concepts:

- Use a while loop menu with these options:
- Basic Arithmetic (use functions + exception handling)
- Scientific Calculations (use math module)
- Generate Random Numbers (use random module)
- Store Results in Dictionary (with timestamp using string)
- View History (show stored results)
- Exit

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS E:\AIML_ITR_2026> & "C:\Program Files\Python314\python.exe"

***** Smart Calculator & Data Manager *****
1. Basic Arithmetic
2. Scientific Calculations
3. Generate Random Numbers
4. Store Last Result In Dictionary
5. View History
6. Exit
Enter Your Choice (1-6):1

*** Basic Arithmetic ***
Operations: + - * /
Choose An Operation (+, -, *, /):+
Enter First Number:1
Enter Second Number:2
Result: 1.0 + 2.0 = 3.0

***** Smart Calculator & Data Manager *****
1. Basic Arithmetic
2. Scientific Calculations
3. Generate Random Numbers
4. Store Last Result In Dictionary
5. View History
6. Exit
Enter Your Choice (1-6):2

*** Scientific Calculations ***
Options: 1) Square Root  2) Power  3) Factorial
Choose An Option (1-3):1
Enter A Number:36
Square Root Of 36.0 = 6.0
```

***** Smart Calculator & Data Manager *****

1. Basic Arithmetic
2. Scientific Calculations
3. Generate Random Numbers
4. Store Last Result In Dictionary
5. View History
6. Exit

Enter Your Choice (1-6):2

*** Scientific Calculations ***

Options: 1) Square Root 2) Power 3) Factorial

Choose An Option (1-3):2

Enter Base:4

Enter Exponent:2

$4.0 \wedge 2.0 = 16.0$

***** Smart Calculator & Data Manager *****

1. Basic Arithmetic
2. Scientific Calculations
3. Generate Random Numbers
4. Store Last Result In Dictionary
5. View History
6. Exit

Enter Your Choice (1-6):2

*** Scientific Calculations ***

Options: 1) Square Root 2) Power 3) Factorial

Choose An Option (1-3):3

Enter A Whole Number:5

Factorial Of 5 = 120

```
***** Smart Calculator & Data Manager *****
```

1. Basic Arithmetic
2. Scientific Calculations
3. Generate Random Numbers
4. Store Last Result In Dictionary
5. View History
6. Exit

```
Enter Your Choice (1-6):3
```

```
*** Generate Random Numbers ***
```

```
How Many Random Numbers Do You Want?3
```

```
Enter The Lower Bound:1
```

```
Enter The Upper Bound:7
```

```
Generated random numbers: [1, 7, 2]
```

```
***** Smart Calculator & Data Manager *****
```

1. Basic Arithmetic
2. Scientific Calculations
3. Generate Random Numbers
4. Store Last Result In Dictionary
5. View History
6. Exit

```
Enter Your Choice (1-6):4
```

```
Result Stored Under Timestamp: 2026-07-20 20:24:32
```

```
***** Smart Calculator & Data Manager *****
```

1. Basic Arithmetic
2. Scientific Calculations
3. Generate Random Numbers
4. Store Last Result In Dictionary
5. View History
6. Exit

```
Enter Your Choice (1-6):5
```

```
*** History ***
```

```
[2026-07-20 20:24:32] Random numbers [1, 7, 2]
```

```
***** Smart Calculator & Data Manager *****
```

1. Basic Arithmetic
2. Scientific Calculations
3. Generate Random Numbers
4. Store Last Result In Dictionary
5. View History
6. Exit

```
Enter Your Choice (1-6):6
```

```
Exiting Smart Calculator & Data Manager. Goodbye!
```

```
PS E:\AIML ITR 2026>
```